

AI DISEASE PREDICTION SYSTEM

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Speaker / Defense Notes

This document provides a detailed explanation of the AI Disease Prediction System, including architecture, methodology, and technical implementation to support project defense.

1. Project Overview

The AI Disease Prediction System is a machine learning-based web application that predicts the most probable disease based on user-selected symptoms. It uses a supervised learning model — Random Forest Classifier — to find patterns between symptoms and diseases.

2. Project Objectives

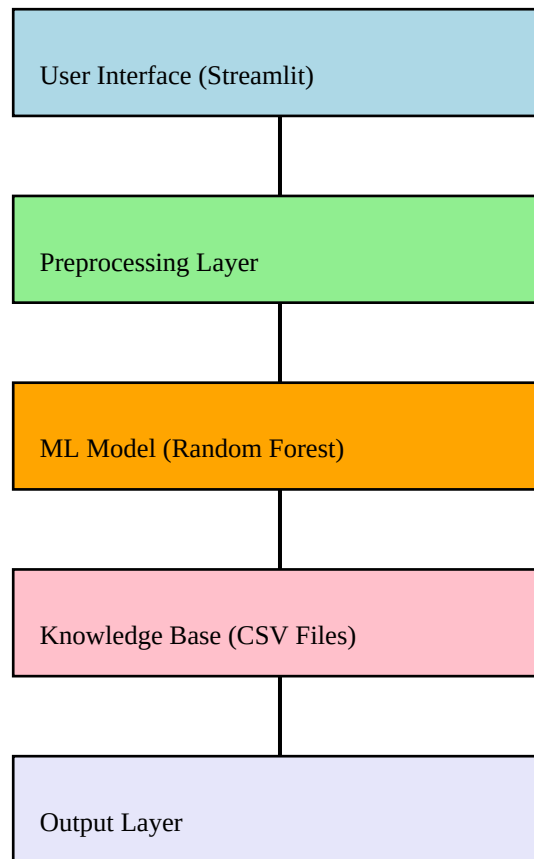
- To develop an intelligent system that predicts diseases using input symptoms.
- To create a user-friendly web app for interactive prediction.
- To encourage early awareness and self-education on possible illnesses.
- To demonstrate AI and ML application in healthcare innovation.

3. Project Scope

This project focuses on disease prediction based solely on symptom data. It does not provide medical diagnosis but rather educational guidance. It is limited to the diseases included in the dataset and uses textual input through a Streamlit web interface.

4. System Architecture and Data Flow

The diagram below illustrates the flow of data and processes within the AI Disease Prediction System, from user input to final prediction output.



****Explanation:**** 1. The user selects symptoms via the Streamlit web interface. 2. The preprocessing layer encodes the symptoms into a binary vector. 3. The trained Random Forest model predicts possible diseases. 4. The knowledge base provides disease descriptions and precautions. 5. Results are displayed back to the user in real time.

5. Technical Implementation

The model training is handled in `train_model.py`, where the dataset is cleaned, features are selected, and the model is trained and saved as `model.pkl`. The web application logic is defined in `app.py`, using Streamlit to interact with users and show prediction results.

6. Model Accuracy and Results

The model achieved 100% accuracy on test data using `RandomForestClassifier`. Predictions are presented alongside confidence percentages and precautionary details.

7. Conclusion

The system effectively demonstrates AI-assisted healthcare education through disease prediction from symptoms. It is reliable, easy to use, and provides valuable health awareness insights. Future extensions may include integration with hospital databases or real-time symptom tracking.

Thank you

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